

INDEPENDENT ENERGY YIELD ASSESSMENT

Hybrid EYA Report

150 MW NHPC FDRE • Tranche-II • Rev. 2



250 MWac

Solar • 375 MWp • 2 sites, Badla



15.75 MW

Wind • 5 × 3.15 MW • Barmer



500 MWh

BESS • 250 MW • 2-hr pooled

SITES	Solar — Badla (2) • Wind — Barmer
GRID	150 MW shared ISTS connectivity • 150 MW contracted FDRE PPA
ASSESSMENT	P50 / P75 / P90 • 20-year horizon



1

Executive Summary

Joulewise Advisory Private Limited has carried out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project (Solar + Wind + BESS) for supply of Firm and Dispatchable RE power under the NHPC 1,200 MW FDRE Tranche-II tender, for a Contracted Capacity of **150 MW** evacuated through a shared 150 MW ISTS connection.

Table 1: Hybrid (Solar + Wind + BESS) Project Details

Project	Site / Location	Total Capacity	Connectivity	GSS
Wind	Barmer region, Rajasthan	15.75 MW (5 x 3.15 MW WTG)	150 MW (shared)	Barmer ISTS GSS
Solar	Badla Site 1, Jodhpur, Rajasthan	200 MWac / 300 MWp (DC:AC 1.50)	150 MW (shared)	Badla ISTS GSS
Solar	Badla Site 2, Jodhpur, Rajasthan	50 MWac / 75 MWp (DC:AC 1.50)	150 MW (shared)	Badla ISTS GSS
BESS	Co-located (pooled)	500 MWh / 250 MW / 2-hr / C-rate 0.50	-	-

Note: The entire 150 MW grid connectivity is shared across the Solar, Wind and BESS components, being the Contracted Capacity under the NHPC FDRE Tranche-II RfS. Only generation meeting the Contracted Capacity is considered for FDRE compliance; energy beyond the evacuation limit is stored in the BESS or curtailed. There is no non-participating / merchant component.

Based on the overall hybrid assessment considering 150 MW PPA supply under the FDRE tender, the following is a summary of results (20-year average) from the hybrid plant including BESS:

Table 2: Hybrid EYA Results Summary (20 Years Average Values)

Sr	Particulars	Basis of Calculation	P50	P75	P90
1	Solar 250 MW Net Gen at 33 kV (GWh)		609.6	589.6	571.5
2	Wind 15.75 MW Net Gen at 33 kV (GWh)		35.9	33.6	31.5
3	Hybrid Net Gen at 33 kV (GWh)	= Sr 1 + Sr 2	645.5	623.2	603.1
4	RE Simultaneous Supply (GWh)		408.6	394.2	381.0
5	RE Supply to ESS (GWh)		146.9	150.7	154.0
6	RE Surplus - Merchant Sales (GWh)		0.0	0.0	0.0
7	RE Curtailment (GWh)	= Sr 3 - Sr 4 - Sr 5	89.9	78.4	68.1
8	RE Transmission Loss (GWh)	= (Sr 4 + Sr 9) x 2.0%	10.7	10.5	10.3
8a	Auxiliary Consumption (GWh)	= (Sr 4 + Sr 9) x 1.5%	8.1	7.9	7.7
9	ESS Supply (GWh)		128.5	131.8	134.7
10	ESS Surplus - Merchant Sales (GWh)		0.0	0.0	0.0
11	ESS RTE Loss (GWh)	= Sr 5 - Sr 9	18.4	18.8	19.3
12	Supply from Other Sources (GWh)		0.0	0.0	0.0
13	Energy Sold under PPA (GWh)	= Sr 4 + Sr 9 - Sr 8 - Sr 8a	518.4	507.6	497.7
14	Annual CUF (%)	= Sr 13 / (150 MW x 8,760 hrs)	39.45	38.63	37.88
15	Shortfall on Annual Availability (MWh)		0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)		0.0	0.0	0.0

Sr	Particulars	Basis of Calculation	P50	P75	P90
17	BESS Usable DC Capacity, Year 1 (MWh)	= Nameplate x SoH-Yr1 x DoD	425.7	425.7	425.7
18	BESS RTE, Year 1 (%)	= Charge Eff x Discharge Eff	87.5	87.5	87.5
19	Total Supply Peak (MW)		150.0	150.0	150.0
20	Total Simultaneous Supply Peak (MW)		150.0	150.0	150.0
21	Total ESS Supply Peak (MW)		150.0	150.0	150.0

Note: Rows shown as 0.0 (RE/ESS Surplus for Merchant Sales, Supply from Other Sources) reflect this assessment's dispatch design, in which generation in excess of the Contracted Capacity that cannot be stored in the BESS is treated as Curtailment rather than a separately dispatchable merchant-sale stream (RfS Clause 6.3 permits such sale; conservatively not quantified here).

2 Introduction

Joulewise Advisory Private Limited has been engaged to carry out an independent Energy Yield Assessment (EYA) of the proposed Hybrid Power Project under the FDRE Tender for 150 MW PPA capacity, consisting of Solar and Wind power projects including a pooled BESS. The Consultant has conducted an independent resource assessment for both the solar and wind portions and performed the hybrid analysis considering the permitted 150 MW power evacuation capacity, tender conditions and PPA requirement. A brief EYA summary of the individual Wind and Solar sites along with the Hybrid project EYA is given in the following sections.

3 Wind EYA Summary

An independent EYA of the proposed wind farm in the Barmer region of Rajasthan has been carried out. The wind farm comprises **5 Wind Turbine Generators (WTGs)** of Suzlon S144 3.15 MW at 140 m hub height, totalling **15.75 MW**.

Table 3: Summary of Wind Resource Assessment (15.75 MW)

Parameter	Value
WTG Model	Suzlon S144_3.15MW (assumed basis)
Rotor Diameter (m)	144
Hub Height (m)	140
WTG Capacity (MW)	3.15
No. of WTGs	5
Project Capacity (MW)	15.75
Total Loss without wake	10.25%
Total Loss with wake	24.35%
Uncertainty (25 years)	9.47%
Net Generation (MWh/MW/year) - P50	2,322.9
Net Generation (MWh/MW/year) - P75	2,134.9
Net Generation (MWh/MW/year) - P90	1,965.6
Net PLF (%) - P50	26.52%
Net PLF (%) - P75	24.37%

Parameter	Value
Net PLF (%) - P90	22.44%

Table 4: Wind Uncertainty Values (S144 3.15 MW at 140 m hub height)

Type of Uncertainty	Uncertainty on AEP (25 Years)
Wind Measurement uncertainty	2.51%
Data Documentation uncertainty	1.00%
Long Term scaling uncertainty	2.00%
Wind resource variability	1.41%
Vertical extrapolation uncertainty	4.00%
Horizontal extrapolation uncertainty	3.75%
Wake modelling	5.50%
Power Curve uncertainty	4.00%
Total	9.47%

Note: The wind resource dataset represents the Barmer site on a per-installed-MW basis (loss cascade consistent with a Suzlon S144 3.15 MW class turbine) and has been scaled to the selected 15.75 MW capacity (5 turbines).

4 **Solar EYA Summary**

The Hybrid Configuration comprises 2 Solar Power projects with AC capacity of 200 MW and 50 MW at Badla (Jodhpur), each co-located with a share of the pooled BESS. Both sites use the same per-MWac PVsyst yield basis (AC CUF 28.91%, referenced to the project's PVsyst simulation), scaled linearly to the selected capacities.

4.1. Site 1 - Badla (200 MWac)

Table 5: Summary of Project & Annual Energy Production (Site 1 - 200 MWac)

Parameter	Value
Project Capacity	300.0 MWp / 200 MWac
DC:AC Ratio	1.50
Location	Badla, District Jodhpur, Rajasthan
Latitude	27.10 N
Longitude	71.30 E
Altitude	210 m
Meteo Dataset	PVsyst Meteonorm (report basis)
GHI (kWh/m2/yr)	1,984.4
Ambient Temp (deg C)	27.23
SPV Technology	Bifacial (Generic BiN, 610 Wp)
Mounting / Tilt	Fixed tilt 29 deg

Parameter	Value
P50 Generation (MWh) - Year 1	506,450
P75 Generation (MWh) - Year 1	489,814
P90 Generation (MWh) - Year 1	474,842
AC CUF (%) - P50	28.91%
DC CUF (%) - P50	19.27%

4.2. Site 2 - Badla (50 MWac)

Table 6: Summary of Project & Annual Energy Production (Site 2 - 50 MWac)

Parameter	Value
Project Capacity	75.0 MWp / 50 MWac
DC:AC Ratio	1.50
Location	Badla, District Jodhpur, Rajasthan
Latitude	27.10 N
Longitude	71.30 E
Altitude	210 m
Meteo Dataset	PVsyst Meteonorm (report basis)
GHI (kWh/m ² /yr)	1,984.4
Ambient Temp (deg C)	27.23
SPV Technology	Bifacial (Generic BiN, 610 Wp)
Mounting / Tilt	Fixed tilt 29 deg
P50 Generation (MWh) - Year 1	126,612
P75 Generation (MWh) - Year 1	122,454
P90 Generation (MWh) - Year 1	118,710
AC CUF (%) - P50	28.91%
DC CUF (%) - P50	19.27%

Note: Site 2 uses the same per-MWac PVsyst yield basis as Site 1, since a single PVsyst reference run underlies both. Equipment counts are indicative and pro-rata; exact selection is subject to detailed engineering.

5 Battery Energy Storage System

A single pooled Battery Energy Storage System (BESS) of **500 MWh / 250 MW (2-hour, C-rate 0.50)** has been considered for hybrid supply, co-located with the solar sites. The Usable DC Capacity applied in the dispatch simulation for each year is derived from the nameplate value via the State-of-Health (SoH) schedule and a Depth-of-Discharge of 95%.

Table 7: Pooled BESS Technical Inputs

Input Parameter	Value
BESS Nameplate / Pol Capacity	500 MWh / 250 MW
Storage Duration	2.0 hours
No. of Containers (indicative)	100 x 5.0 MWh
Charge Efficiency	92.0%
Discharge Efficiency	92.0%
Round Trip Efficiency (RTE), Year 1	87.77%
Cycles per Day	1
DoD of usable capacity	95%
Battery Augmentation	100% SoH restored, Year 11
Usable Capacity Formula	Nameplate (MWh) x SoH (%) x DoD (%)
Auxiliary Consumption	1.5% of Gross Supply/year

Table 8: Pooled BESS Usable DC Capacity and RTE Schedule

Yr	DC Usable (MWh)	SoH (%)	RTE (%)
1	475.0	100.00%	87.77%
2	456.4	96.08%	87.64%
3	446.8	94.07%	87.58%
4	438.9	92.40%	87.54%
5	431.8	90.91%	87.51%
6	425.3	89.54%	87.48%
7	419.3	88.27%	87.46%
8	413.6	87.07%	87.43%
9	408.2	85.93%	87.41%
10	403.0	84.84%	87.39%
11	475.0	100.00%	87.77%
12	456.4	96.08%	87.64%
13	446.8	94.07%	87.58%
14	438.9	92.40%	87.54%
15	431.8	90.91%	87.51%
16	425.3	89.54%	87.48%
17	419.3	88.27%	87.46%
18	413.6	87.07%	87.43%
19	408.2	85.93%	87.41%
20	403.0	84.84%	87.39%



6.1. Methodology

A rule-based, chronological hourly dispatch simulation models the hybrid plant's generation, BESS charge/discharge behaviour and settlement against the tender's Firm and Dispatchable RE supply conditions. The objective maximises revenue-generating energy = Energy Supplied under PPA – 1.5 × Energy Shortfall + 0.5 × Energy Surplus, subject to annual CUF band (–15% / +10% of declared CUF), monthly 90% peak availability, and the 150 MW evacuation limit in every time block.

INPUTS

- ◆ Solar generation scaled from the PVsyst per-MWac yield to 250 MWac, degraded 0.40%/year.
- ◆ Wind generation scaled from the per-MW resource basis to 15.75 MW, degraded 0.20%/year.
- ◆ Pooled BESS: year-specific Usable Capacity (Nameplate x SoH x DoD) and Round-Trip Efficiency; 100% SoH augmentation in Year 11.
- ◆ Transmission loss of 2.0% and Auxiliary Consumption of 1.5% deducted from (Simultaneous + ESS) supply to reach Net Supply under the PPA.

6.2. Hybrid Project Results

The following table provides the 20-year average generation values used for the assessment (reproduces Table 2).

Table 9: Hybrid EYA Results Summary (20 Years Average Values)

Sr	Particulars	Basis of Calculation	P50	P75	P90
1	Solar 250 MW Net Gen at 33 kV (GWh)		609.6	589.6	571.5
2	Wind 15.75 MW Net Gen at 33 kV (GWh)		35.9	33.6	31.5
3	Hybrid Net Gen at 33 kV (GWh)	= Sr 1 + Sr 2	645.5	623.2	603.1
4	RE Simultaneous Supply (GWh)		408.6	394.2	381.0
5	RE Supply to ESS (GWh)		146.9	150.7	154.0
6	RE Surplus - Merchant Sales (GWh)		0.0	0.0	0.0
7	RE Curtailment (GWh)	= Sr 3 - Sr 4 - Sr 5	89.9	78.4	68.1
8	RE Transmission Loss (GWh)	= (Sr 4 + Sr 9) x 2.0%	10.7	10.5	10.3
8a	Auxiliary Consumption (GWh)	= (Sr 4 + Sr 9) x 1.5%	8.1	7.9	7.7
9	ESS Supply (GWh)		128.5	131.8	134.7
10	ESS Surplus - Merchant Sales (GWh)		0.0	0.0	0.0
11	ESS RTE Loss (GWh)	= Sr 5 - Sr 9	18.4	18.8	19.3
12	Supply from Other Sources (GWh)		0.0	0.0	0.0
13	Energy Sold under PPA (GWh)	= Sr 4 + Sr 9 - Sr 8 - Sr 8a	518.4	507.6	497.7
14	Annual CUF (%)	= Sr 13 / (150 MW x 8,760 hrs)	39.45	38.63	37.88
15	Shortfall on Annual Availability (MWh)		0.0	0.0	0.0
16	Shortfall on Monthly Availability (MWh)		0.0	0.0	0.0
17	BESS Usable DC Capacity, Year 1 (MWh)	= Nameplate x SoH-Yr1 x DoD	425.7	425.7	425.7

Sr	Particulars	Basis of Calculation	P50	P75	P90
18	BESS RTE, Year 1 (%)	= Charge Eff x Discharge Eff	87.5	87.5	87.5
19	Total Supply Peak (MW)		150.0	150.0	150.0
20	Total Simultaneous Supply Peak (MW)		150.0	150.0	150.0
21	Total ESS Supply Peak (MW)		150.0	150.0	150.0

Table 10: P50 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply	ESS Charge-In	ESS RTE Loss	ESS Supply	Curtailment	Tx Loss	Aux Consum.	Gross Supply	Net PPA Supply	Req. Peak	Actual Peak	Peak Avail (%)
January	51,475	30,458	13,648	1,647	12,001	7,775	849	637	42,458	40,972	18,600	18,600	100.0
February	50,833	30,940	11,482	1,414	10,068	8,692	820	615	41,008	39,573	16,800	16,800	100.0
March	63,563	38,524	11,533	1,407	10,126	13,801	973	730	48,650	46,947	18,600	18,600	100.0
April	64,494	40,574	10,393	1,313	9,080	13,813	993	745	49,653	47,915	18,000	18,000	100.0
May	70,377	45,026	9,858	1,199	8,659	15,789	1,074	805	53,684	51,805	18,600	18,600	100.0
June	54,563	37,045	10,915	1,317	9,598	6,962	933	700	46,643	45,010	18,000	18,000	100.0
July	51,108	35,013	12,260	1,518	10,742	4,321	915	686	45,755	44,154	18,600	18,600	100.0
August	51,680	34,736	12,444	1,556	10,888	4,952	912	684	45,624	44,027	18,600	18,600	100.0
September	50,338	32,227	12,433	1,441	10,992	6,115	864	648	43,219	41,707	18,000	18,000	100.0
October	58,207	35,207	12,380	1,527	10,852	10,926	921	691	46,059	44,447	18,600	18,600	100.0
November	53,083	31,930	12,608	1,533	11,075	8,862	860	645	43,005	41,500	18,000	18,000	100.0
December	49,923	29,107	14,153	1,753	12,401	7,165	830	623	41,508	40,055	18,600	18,600	100.0
ANNUAL TOTAL	669,644	420,785	144,106	17,624	126,482	109,172	10,945	8,209	547,268	528,113	219,000	219,000	

Table 11: P75 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply	ESS Charge-In	ESS RTE Loss	ESS Supply	Curtailment	Tx Loss	Aux Consum.	Gross Supply	Net PPA Supply	Req. Peak	Actual Peak	Peak Avail (%)
January	49,610	29,209	14,043	1,698	12,345	6,848	831	623	41,554	40,100	18,600	18,600	100.0
February	49,256	30,050	11,699	1,437	10,262	7,819	806	605	40,311	38,901	16,800	16,800	100.0
March	61,970	37,668	11,752	1,436	10,316	12,845	960	720	47,984	46,304	18,600	18,600	100.0
April	62,825	39,337	10,604	1,336	9,268	13,169	972	729	48,605	46,904	18,000	18,000	100.0
May	68,492	44,141	10,039	1,223	8,815	14,607	1,059	794	52,956	51,103	18,600	18,600	100.0
June	52,175	35,636	11,266	1,359	9,907	5,672	911	683	45,543	43,949	18,000	18,000	100.0
July	48,822	33,429	12,671	1,562	11,110	3,247	891	668	44,538	42,980	18,600	18,600	100.0
August	49,345	33,135	12,881	1,616	11,265	3,837	888	666	44,400	42,846	18,600	18,600	100.0
September	48,196	30,794	12,796	1,487	11,309	5,086	842	632	42,103	40,630	18,000	18,000	100.0
October	56,388	34,526	12,623	1,557	11,066	9,568	912	684	45,592	43,996	18,600	18,600	100.0
November	51,440	30,907	12,869	1,559	11,309	8,028	844	633	42,216	40,738	18,000	18,000	100.0
December	47,999	27,802	14,545	1,804	12,741	6,207	811	608	40,543	39,124	18,600	18,600	100.0
ANNUAL TOTAL	646,519	406,634	147,787	18,074	129,713	96,932	10,727	8,045	536,347	517,575	219,000	219,000	

Table 12: P90 Monthly Energy Settlement (Year 1)

Month	RE Gen (MWh)	Simult. Supply	ESS Charge-In	ESS RTE Loss	ESS Supply	Curtailment	Tx Loss	Aux Consum.	Gross Supply	Net PPA Supply	Req. Peak	Actual Peak	Peak Avail (%)
January	47,897	28,141	14,383	1,740	12,643	5,916	816	612	40,784	39,357	18,600	18,600	100.0
February	47,751	29,141	11,923	1,459	10,464	7,036	792	594	39,605	38,218	16,800	16,800	100.0
March	60,477	36,991	11,941	1,465	10,475	11,839	949	712	47,467	45,805	18,600	18,600	100.0
April	61,267	38,302	10,802	1,358	9,444	12,448	955	716	47,746	46,075	18,000	18,000	100.0
May	66,737	43,125	10,241	1,249	8,992	13,666	1,042	782	52,117	50,293	18,600	18,600	100.0
June	50,132	34,320	11,611	1,403	10,208	4,644	891	668	44,528	42,970	18,000	18,000	100.0
July	46,880	31,900	13,052	1,603	11,450	2,495	867	650	43,350	41,832	18,600	18,600	100.0
August	47,371	31,710	13,250	1,662	11,588	2,954	866	649	43,297	41,782	18,600	18,600	100.0
September	46,335	29,597	13,080	1,526	11,554	4,143	823	617	41,151	39,710	18,000	18,000	100.0
October	54,707	33,741	12,837	1,583	11,254	8,479	900	675	44,995	43,420	18,600	18,600	100.0
November	49,885	29,829	13,139	1,590	11,550	7,327	828	621	41,378	39,930	18,000	18,000	100.0
December	46,260	26,811	14,860	1,845	13,016	5,167	797	597	39,827	38,433	18,600	18,600	100.0
ANNUAL TOTAL	625,701	393,607	151,120	18,482	132,638	86,115	10,525	7,894	526,245	507,827	219,000	219,000	

 Annexure 1: Yearly Results

P50 - Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 250 MW Net Gen (GWh)	633.1	630.5	628.0	625.5	623.0	620.5	618.0	615.6	613.1	610.6
Wind 15.75 MW Net Gen (GWh)	36.6	36.5	36.4	36.4	36.3	36.2	36.2	36.1	36.0	35.9
Hybrid Net Gen (GWh)	669.6	667.1	664.5	661.9	659.3	656.7	654.2	651.7	649.1	646.6
RE Simultaneous Supply (GWh)	420.8	419.5	418.2	416.9	415.6	414.3	413.1	411.8	410.5	409.3
RE Supply to ESS (GWh)	144.1	144.4	144.7	145.0	145.3	145.6	145.8	146.1	146.4	146.7
RE Curtailment (GWh)	104.8	103.2	101.6	100.0	98.4	96.9	95.3	93.7	92.2	90.6
RE Transmission Loss (GWh)	10.9	10.9	10.9	10.9	10.9	10.8	10.8	10.8	10.8	10.7
Auxiliary Consumption (GWh)	8.2	8.2	8.2	8.2	8.1	8.1	8.1	8.1	8.1	8.1
ESS Supply (GWh)	126.5	126.6	126.8	127.0	127.1	127.3	127.6	127.8	128.0	128.2
ESS RTE Loss (GWh)	17.6	17.8	17.9	18.0	18.1	18.2	18.3	18.4	18.4	18.5
Energy Sold under PPA (GWh)	528.1	527.0	525.9	524.8	523.7	522.7	521.7	520.7	519.7	518.6
Annual CUF (%)	40.19	40.11	40.02	39.94	39.86	39.78	39.70	39.63	39.55	39.47
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	475.0	464.2	453.4	442.6	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.77	87.71	87.64	87.58	87.51	87.49	87.46	87.44	87.41	87.39

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 250 MW Net Gen (GWh)	608.2	605.8	603.4	600.9	598.5	596.1	593.8	591.4	589.0	586.6
Wind 15.75 MW Net Gen (GWh)	35.9	35.8	35.7	35.6	35.6	35.5	35.4	35.4	35.3	35.2
Hybrid Net Gen (GWh)	644.1	641.6	639.1	636.6	634.1	631.6	629.2	626.8	624.3	621.9
RE Simultaneous Supply (GWh)	408.0	406.8	405.5	404.2	403.0	401.7	400.4	399.0	397.7	396.4
RE Supply to ESS (GWh)	147.0	147.4	147.7	148.0	148.3	148.6	148.9	149.2	149.5	149.7
RE Curtailment (GWh)	89.0	87.5	85.9	84.3	82.8	81.3	79.9	78.5	77.1	75.7
RE Transmission Loss (GWh)	10.7	10.7	10.7	10.7	10.7	10.6	10.6	10.6	10.6	10.5
Auxiliary Consumption (GWh)	8.0	8.0	8.0	8.0	8.0	8.0	8.0	7.9	7.9	7.9
ESS Supply (GWh)	128.5	128.8	129.2	129.5	129.8	130.0	130.2	130.4	130.7	130.9
ESS RTE Loss (GWh)	18.5	18.5	18.5	18.5	18.5	18.6	18.7	18.7	18.8	18.9
Energy Sold under PPA (GWh)	517.7	516.8	516.0	515.1	514.2	513.1	512.0	510.9	509.9	508.8
Annual CUF (%)	39.40	39.33	39.27	39.20	39.13	39.05	38.97	38.89	38.80	38.72
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	408.8	414.5	420.3	426.1	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.41	87.44	87.46	87.49	87.51	87.49	87.46	87.44	87.41	87.39

Monthly Shortage (20-year average, MWh)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
20-yr Avg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

P75 - Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 250 MW Net Gen (GWh)	612.3	609.8	607.4	605.0	602.5	600.1	597.7	595.4	593.0	590.6
Wind 15.75 MW Net Gen (GWh)	34.3	34.2	34.1	34.0	34.0	33.9	33.8	33.8	33.7	33.6
Hybrid Net Gen (GWh)	646.5	644.0	641.5	639.0	636.5	634.0	631.6	629.1	626.7	624.2
RE Simultaneous Supply (GWh)	406.6	405.3	404.0	402.6	401.3	400.0	398.7	397.4	396.0	394.7
RE Supply to ESS (GWh)	147.8	148.1	148.4	148.7	149.0	149.3	149.5	149.8	150.1	150.4
RE Curtailment (GWh)	92.1	90.6	89.2	87.7	86.2	84.8	83.4	82.0	80.5	79.1
RE Transmission Loss (GWh)	10.7	10.7	10.7	10.7	10.6	10.6	10.6	10.6	10.5	10.5
Auxiliary Consumption (GWh)	8.0	8.0	8.0	8.0	8.0	8.0	7.9	7.9	7.9	7.9
ESS Supply (GWh)	129.7	129.9	130.0	130.2	130.4	130.6	130.8	131.0	131.2	131.4
ESS RTE Loss (GWh)	18.1	18.2	18.3	18.5	18.6	18.7	18.8	18.8	18.9	19.0
Energy Sold under PPA (GWh)	517.6	516.4	515.3	514.2	513.1	512.0	510.9	509.9	508.8	507.7
Annual CUF (%)	39.39	39.30	39.22	39.13	39.05	38.96	38.88	38.80	38.72	38.64
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	475.0	464.2	453.4	442.6	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.77	87.71	87.64	87.58	87.51	87.49	87.46	87.44	87.41	87.39

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 250 MW Net Gen (GWh)	588.2	585.9	583.5	581.2	578.9	576.6	574.3	572.0	569.7	567.4
Wind 15.75 MW Net Gen (GWh)	33.6	33.5	33.4	33.4	33.3	33.2	33.2	33.1	33.0	33.0
Hybrid Net Gen (GWh)	621.8	619.4	617.0	614.6	612.2	609.8	607.4	605.1	602.7	600.3
RE Simultaneous Supply (GWh)	393.5	392.2	390.9	389.6	388.3	387.0	385.7	384.4	383.1	381.8
RE Supply to ESS (GWh)	150.7	151.1	151.4	151.8	152.2	152.4	152.7	153.0	153.2	153.5
RE Curtailment (GWh)	77.6	76.1	74.6	73.1	71.7	70.3	69.0	67.7	66.4	65.0
RE Transmission Loss (GWh)	10.5	10.5	10.5	10.4	10.4	10.4	10.4	10.4	10.3	10.3
Auxiliary Consumption (GWh)	7.9	7.9	7.9	7.8	7.8	7.8	7.8	7.8	7.8	7.7
ESS Supply (GWh)	131.8	132.1	132.5	132.8	133.2	133.4	133.6	133.8	134.0	134.2
ESS RTE Loss (GWh)	19.0	19.0	19.0	19.0	19.0	19.1	19.1	19.2	19.3	19.4
Energy Sold under PPA (GWh)	506.8	505.9	505.0	504.1	503.2	502.2	501.1	500.0	499.0	497.9
Annual CUF (%)	38.57	38.50	38.44	38.37	38.30	38.22	38.14	38.05	37.97	37.89
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	408.8	414.5	420.3	426.1	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.41	87.44	87.46	87.49	87.51	87.49	87.46	87.44	87.41	87.39

Monthly Shortage (20-year average, MWh)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
20-yr Avg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

P90 - Yearly Results (20 Years)

Years 1-10

Particulars	Yr 1	Yr 2	Yr 3	Yr 4	Yr 5	Yr 6	Yr 7	Yr 8	Yr 9	Yr 10
Solar 250 MW Net Gen (GWh)	593.6	591.2	588.8	586.5	584.1	581.8	579.5	577.2	574.8	572.5
Wind 15.75 MW Net Gen (GWh)	32.1	32.1	32.0	32.0	31.9	31.8	31.8	31.7	31.6	31.6
Hybrid Net Gen (GWh)	625.7	623.3	620.9	618.4	616.0	613.6	611.2	608.9	606.5	604.1
RE Simultaneous Supply (GWh)	393.6	392.2	390.9	389.5	388.1	386.8	385.5	384.3	383.0	381.7
RE Supply to ESS (GWh)	151.1	151.4	151.7	152.0	152.3	152.6	152.9	153.1	153.4	153.7
RE Curtailment (GWh)	81.0	79.6	78.3	76.9	75.5	74.2	72.8	71.5	70.1	68.8
RE Transmission Loss (GWh)	10.5	10.5	10.5	10.5	10.4	10.4	10.4	10.4	10.3	10.3
Auxiliary Consumption (GWh)	7.9	7.9	7.9	7.8	7.8	7.8	7.8	7.8	7.8	7.7
ESS Supply (GWh)	132.6	132.8	133.0	133.1	133.3	133.5	133.7	133.9	134.1	134.3
ESS RTE Loss (GWh)	18.5	18.6	18.8	18.9	19.0	19.1	19.2	19.2	19.3	19.4
Energy Sold under PPA (GWh)	507.8	506.7	505.5	504.4	503.2	502.1	501.1	500.0	499.0	497.9
Annual CUF (%)	38.65	38.56	38.47	38.38	38.29	38.21	38.13	38.05	37.97	37.89
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	475.0	464.2	453.4	442.6	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.77	87.71	87.64	87.58	87.51	87.49	87.46	87.44	87.41	87.39

Years 11-20

Particulars	Yr 11	Yr 12	Yr 13	Yr 14	Yr 15	Yr 16	Yr 17	Yr 18	Yr 19	Yr 20
Solar 250 MW Net Gen (GWh)	570.3	568.0	565.7	563.4	561.2	558.9	556.7	554.5	552.3	550.0
Wind 15.75 MW Net Gen (GWh)	31.5	31.4	31.4	31.3	31.3	31.2	31.1	31.1	31.0	30.9
Hybrid Net Gen (GWh)	601.8	599.4	597.1	594.8	592.4	590.1	587.8	585.6	583.3	581.0
RE Simultaneous Supply (GWh)	380.4	379.1	377.8	376.5	375.2	373.9	372.5	371.2	369.9	368.6
RE Supply to ESS (GWh)	154.1	154.4	154.8	155.2	155.6	155.8	156.1	156.3	156.6	156.9
RE Curtailment (GWh)	67.3	65.9	64.5	63.1	61.7	60.5	59.2	58.0	56.8	55.6
RE Transmission Loss (GWh)	10.3	10.3	10.3	10.2	10.2	10.2	10.2	10.2	10.1	10.1
Auxiliary Consumption (GWh)	7.7	7.7	7.7	7.7	7.7	7.7	7.6	7.6	7.6	7.6
ESS Supply (GWh)	134.7	135.0	135.4	135.8	136.1	136.3	136.5	136.7	136.9	137.1
ESS RTE Loss (GWh)	19.4	19.4	19.4	19.4	19.4	19.5	19.6	19.6	19.7	19.8
Energy Sold under PPA (GWh)	497.0	496.1	495.2	494.3	493.4	492.3	491.2	490.1	489.0	487.9
Annual CUF (%)	37.82	37.76	37.69	37.62	37.55	37.47	37.38	37.30	37.22	37.13
Shortfall on Monthly Availability (MWh)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
BESS Usable DC Capacity (MWh)	408.8	414.5	420.3	426.1	431.8	426.1	420.3	414.5	408.8	403.0
BESS RTE (%)	87.41	87.44	87.46	87.49	87.51	87.49	87.46	87.44	87.41	87.39

Monthly Shortage (20-year average, MWh)

Metric	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
20-yr Avg	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Note: Wind generation was estimated at the interconnection level (including line/transformer losses) and adjusted to the 33 kV level; solar generation was estimated directly at 33 kV. A rule-based chronological dispatch determined BESS charging and the overall hybrid energy. Dispatch-dependent quantities (simultaneous vs BESS split, curtailment, peak availability, shortfall) are computed by the FDRE dispatch engine for the 150 MW contracted capacity; generation is scaled from the report's per-MW PVsyst/wind bases. Figures are screening-grade.